CS4168 Data Mining Project

Submitted by Malleshwari Kurva — 25252704

Neha Vinod — 25367331

Ganga Bhavani Chintalapalli — 25303031

Lekha Murthy — *25232363*

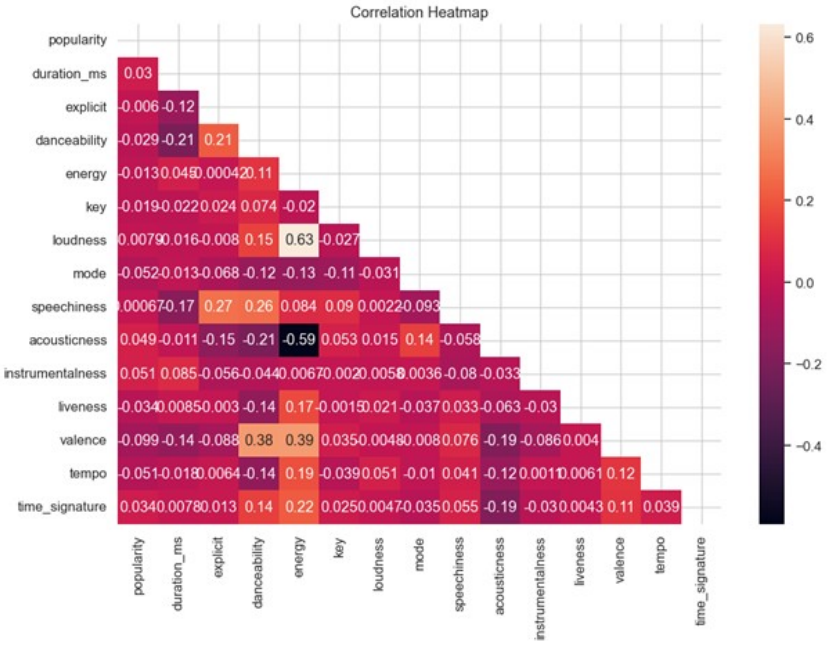
Yaswitha Vasipalli — 25290703

Project Overview

Team Member	Responsibility
Malleshwari Kurva	Data preprocessing, feature scaling, encoding, and pipeline setup
Neha Vinod	EDA, visualisation and pattern analysis
Ganga Bhavani Chintalapalli	Clustering (k-means and DBSCAN),tuning, and evaluation
Lekha Murthy	Classification modelling and model selection
Yaswitha Vasipalli	Regression modelling and result analysis

- Goal: Understand and predict popularity
- Dataset: tracks2026.csv

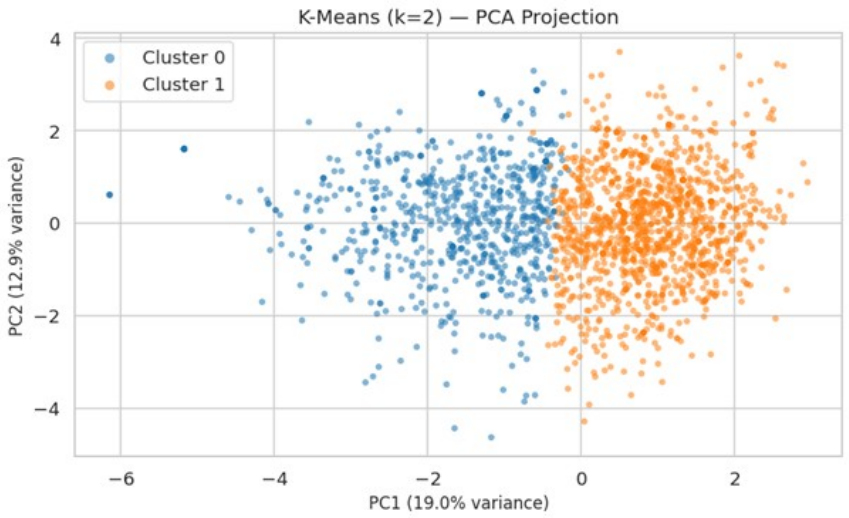
EDA Highlights

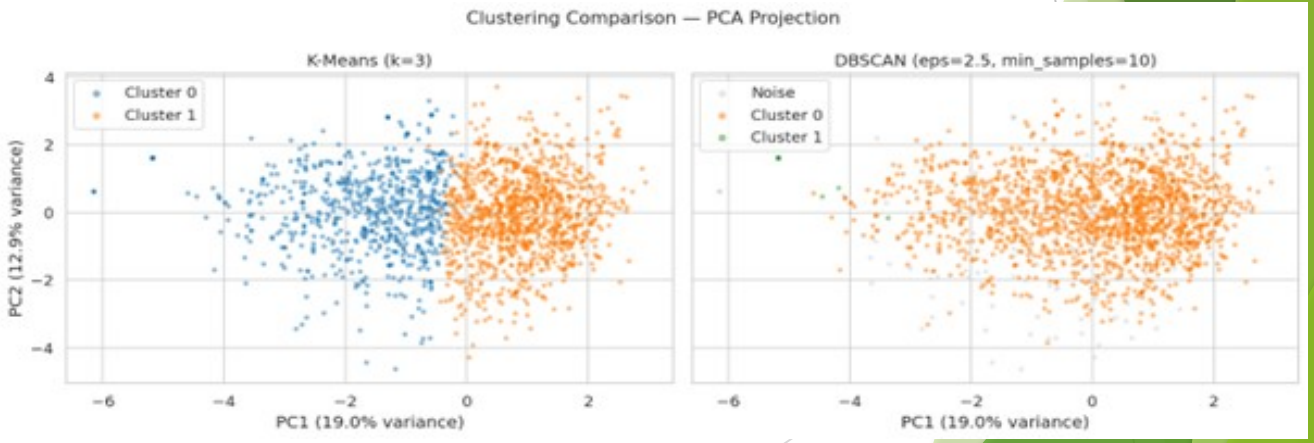


- Explored distributions, correlations, and genre patterns.
- Identified features like energy, loudness, and danceability show moderate positive correlation with popularity.
- Acousticness and instrumentalness tend to have negative relationships with popularity
- Significant feature variance across genres, suggesting clustering may work partially

Clustering (k-means & DBSCAN)

- Applied clustering after dropping track_genre.
- Compared multiple k values.
- k-means produced **partially separable clusters**, but with noticeable overlap.
- DBSCAN struggled due to **high dimensionality and density variation**, leading to noise points
- Final choice: **k-means as it is the** best trade-off between structure and interpretability





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============================================================
MODEL: Logistic Regression
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Accuracy : 0.6140
F1 Score : 0.6321
ROC-AUC  : 0.6853
------------------------------------------------------------
Classification Report:
------------------------------------------------------------
              precision  recall  f1-score  support
0                 0.645   0.551     0.594  198.000
1                 0.590   0.681     0.632  188.000
accuracy          0.614   0.614     0.614    0.614
macro avg         0.617   0.616     0.613  386.000
weighted avg      0.618   0.614     0.613  386.000
============================================================


============================================================
MODEL: Random Forest
============================================================
Accuracy : 0.7383
F1 Score : 0.7335
ROC-AUC  : 0.7980
------------------------------------------------------------
...
accuracy          0.668   0.668     0.668    0.668
macro avg         0.669   0.667     0.667  386.000
weighted avg      0.669   0.668     0.667  386.000
============================================================
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Classification Task

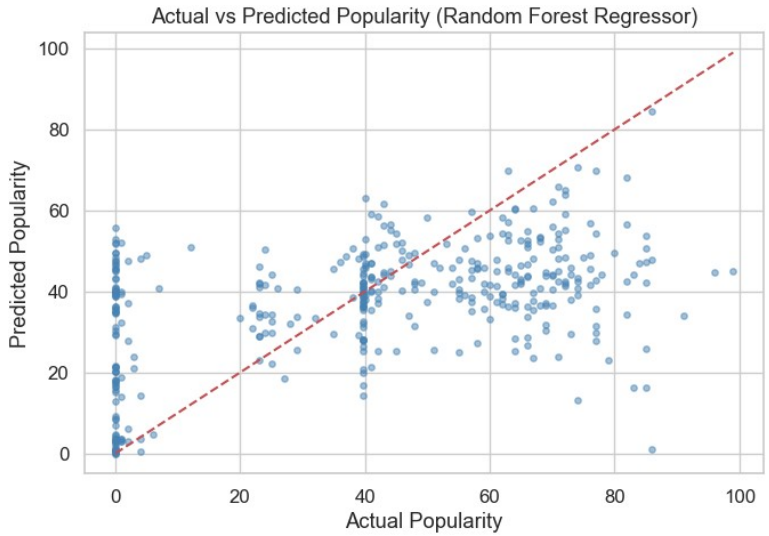
- Binary classification (high vs low popularity) achieved **good performance**
- Models tested: (e.g., Logistic Regression, Random Forest, Decision Tree)
- Logistic Regression provided a strong baseline but is limited in capturing non-linear relationships. Decision Trees showed slightly lower generalization performance due to overfitting tendencies.
- **Random Forest** consistently achieved the highest F1-score and ROC-AUC, indicating better balance between precision and recall.

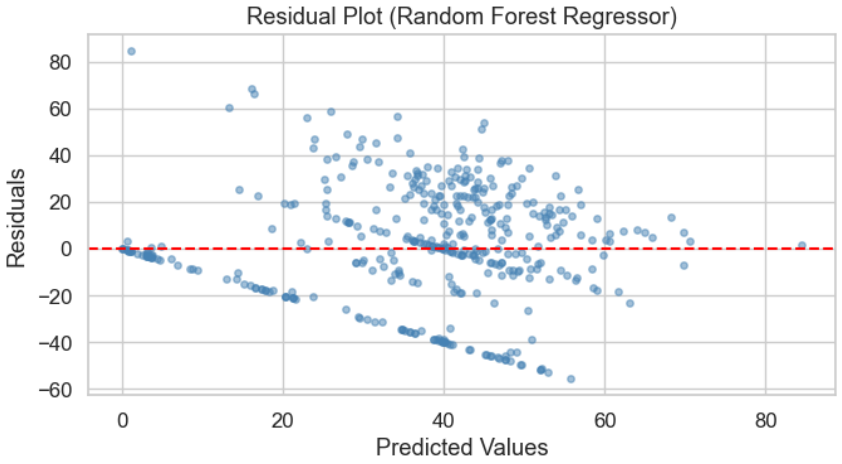
Feature Importance:

- The results show that **acousticness, duration, tempo, and speechiness** are among the most important predictors, followed closely by **energy and valence**.

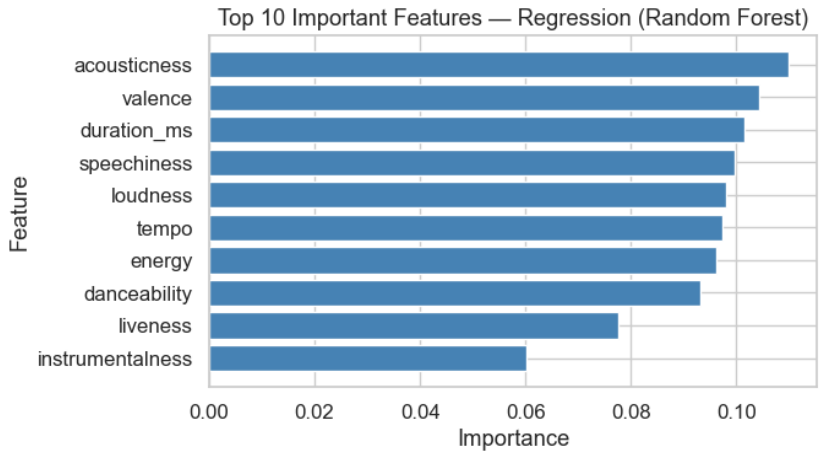
Regression Results

- Random Forest performed best with the lowest error and highest R^2.
- The Actual vs Predicted plot shows strong agreement between values. Residuals are evenly spread around zero, indicating no major bias.
- Overall, audio features help predict popularity, but noise makes exact prediction difficult.





Key Discovery



- Popularity is influenced more by audio characteristics than genre labels
- High-energy, loud, and danceable tracks are more likely to succeed
- Genre alone does not fully explain clusters.

Modelling Decision

- One modelling decision that improved performance was using feature scaling and proper pipelines, which ensured consistent preprocessing and improved model stability.
- One decision that reduced performance was including all features without selection, which introduced noise and reduced accuracy, especially for DBSCAN and regression models.

Conclusion

- Audio features such as energy, loudness, and danceability strongly influence music popularity
- Clustering revealed partial structure, but did not fully align with genre labels
- Classification performed better than regression for predicting popularity
- Random Forest was the best-performing model across tasks due to handling non-linear relationships effectively
- Predicting exact popularity is challenging due to noise and missing external factors (e.g., trends, artist fame)
- Overall, audio features provide useful insights, but are not sufficient for perfect prediction